Perturbations of Spherically Symmetric Spacetimes in $f(R) = R^2$ Gravity: Personal Notes

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1 Field Equations in Jordan Frame

Varying the action with respect to the metric $g_{\mu\nu}$, the field equations for general $f(R)$ gravity in vacuum are:

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)\, f'(R) = 0 \tag{1}$$

where:

- $f'(R) = \frac{df(R)}{dR}$,
- $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$

Contracting with the inverse metric $g^{\mu\nu}$, we get the trace equation:

$$f'(R)R - 2f(R) + 3\Box f'(R) = 0 \tag{2}$$

With our model $f(R) = R^2$, $f'(R) = 2R$, so the field equations and the trace equation look like:

$$2RR_{\mu\nu} - \frac{1}{2}R^2 g_{\mu\nu} + 2(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)R = 0 \tag{3}$$

and

$$6\Box R = 0 \implies \Box R = 0 \tag{4}$$

This is a massless Klein-Gordon equation for the Ricci scalar R.

2 Can we use Einstein frame for $f(R) = R^2$?

we will study if the action of the pure R^2 gravity can be studied in the Einstein frame.

$$S_{R^2} = \int d^4x\sqrt{-g}\beta R^2 \tag{5}$$

Following the analysis of Alvarez-Gaume et al. (2016), let us derive the corresponding action in the Einstein frame. For this, let us consider the following action:

$$S = \int d^4x\sqrt{-g}\left(\phi R - \frac{1}{4\beta}\phi^2\right). \tag{6}$$

Here, ϕ satisfies the constraint:

$$\phi = 2\beta R. \tag{7}$$

By substituting it into (6), it can easily be shown that the action reduces to the original one, given by (5). Furthermore, let us define a conformally rescaled metric:

$$\tilde{g}_{\mu\nu} = \frac{2}{M_{\mathrm{P}}^2}\phi g_{\mu\nu} \tag{8}$$

By expressing (6) in terms of the new metric, we find:

$$S_{(E)} = \int d^4x\sqrt{-\tilde{g}}\left[\frac{M_{\mathrm{P}}^2}{2}\left(\tilde{R} - 2\Lambda_\beta\right) - \frac{1}{2}\tilde{g}^{\mu\nu}\partial_\mu\Phi_n\partial_\nu\Phi_n\right], \tag{9}$$

where $\tilde{R}$ is the curvature corresponding to the new metric, and

$$\Lambda_\beta = \frac{M_\mathrm{P}^2}{16\beta}, \quad \Phi_n = \sqrt{\frac{3}{2}} M_\mathrm{P} \ln\left(\frac{2\phi}{M_\mathrm{P}^2}\right). \tag{10}$$

The action (9) is the action of the pure R^2 gravity in the Einstein frame. As pointed out in Alvarez-Gaume et al. (2016), this formulation only works for $R \neq 0$. In the (Ricci) flat case, $R = 0$, and thus we have $\phi = 0$. As a result, the conformally rescaled metric $\tilde{g}_{\mu\nu}$ is not defined, and we cannot use the Einstein frame in order to describe R^2 gravity in (Ricci) flat spacetime.

We could also intuitively understand why the connection between the original and Einstein frame is absent for the flat-space metric $g_{\mu\nu} = \eta_{\mu\nu}$ – the action (8) has cosmological constant, and thus the flat space is not even the solution of the equations for the metric $\tilde{g}_{\mu\nu}$.

3 Linearization of R^2 Gravity Around Schwarzschild Background in the Jordan Frame

3.1 Introduction

In this section, we study the linearized perturbations of $f(R) = R^2$ gravity about a Schwarzschild background, working in the Jordan frame. The aim is to derive the linearized field equations satisfied by small perturbations around Schwarzschild spacetime, without transforming to the Einstein frame.

Unlike General Relativity, where the linearized equations around Schwarzschild involve only metric perturbations, in R^2 gravity, the theory has an additional propagating scalar degree of freedom associated with the Ricci scalar R. Understanding the behavior of these perturbations is essential for computing quasi-normal modes (QNMs) in R^2 gravity.

3.2 Linearization Around Schwarzschild

The action for R^2 gravity in the Jordan frame is given by (5). Varying the action with respect to $g_{\mu\nu}$ yields the field equations (1). We consider perturbations around a Schwarzschild background, whose metric is:

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2\left(d\theta^2 + \sin^2\theta\, d\phi^2\right), \tag{11}$$

with

$$f(r) = 1 - \frac{2M}{r}.$$

The background Ricci tensor and Ricci scalar satisfy:

$$R_{\mu\nu}^{(0)} = 0, \qquad R^{(0)} = 0, \tag{12}$$

where the superscript (0) denotes background quantities.

We introduce small perturbations to the metric:

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + h_{\mu\nu}, \tag{13}$$

and correspondingly, the Ricci scalar perturbation is:

$$R = \delta R, \tag{14}$$

since $R^{(0)} = 0$.

Expanding the field equations to linear order, we analyze each term separately:

- The term $2RR_{\mu\nu}$ becomes zero at linear order because both $R^{(0)} = 0$ and $R^{(0)}_{\mu\nu} = 0$.
- The term $-\frac{1}{2}R^2 g_{\mu\nu}$ is quadratic in δR, and hence vanishes at linear order.
- The term $-2\nabla_\mu \nabla_\nu R$ linearizes to:

$$-2\nabla_\mu \nabla_\nu \delta R. \tag{15}$$

- The term $+2g_{\mu\nu}\Box R$ linearizes to:

$$2g^{(0)}_{\mu\nu}\Box\delta R, \tag{16}$$

  where $\Box$ the d'Alembertian.

Therefore, the linearized field equations become:

$$\nabla_\mu \nabla_\nu \delta R = g^{(0)}_{\mu\nu}\Box\delta R. \tag{17}$$

Taking the trace of this equation with $g^{(0)\mu\nu}$, we find:

$$g^{(0)\mu\nu}\nabla^{(0)}_\mu \nabla^{(0)}_\nu \delta R = g^{(0)\mu\nu} g^{(0)}_{\mu\nu}\Box\delta R. \tag{18}$$

Since $g^{(0)\mu\nu} g^{(0)}_{\mu\nu} = 4$ in four spacetime dimensions, this simplifies to:

$$\Box\delta R = 4\Box\delta R. \tag{19}$$

Rearranging, we obtain:

$$\Box\delta R = 0. \tag{20}$$

Thus, δR satisfies the massless scalar wave equation on the Schwarzschild background.

3.3 Summary

In the Jordan frame, the linearization of R^2 gravity around Schwarzschild background leads to a significant simplification: the perturbation of the Ricci scalar, δR, obeys the massless Klein-Gordon equation on the Schwarzschild spacetime. Therefore, the analysis of scalar-type quasi-normal modes in R^2 gravity reduces to studying the behavior of a massless scalar field in the Schwarzschild geometry. The study of metric perturbations $h_{\mu\nu}$ and their coupling to δR may introduce additional complexities, which can be explored in further sections.

4 Scalar Perturbations: Solving the Massless Scalar Wave Equation in Schwarzschild Background

We now solve the massless scalar wave equation for the Ricci scalar perturbation δR in the Schwarzschild background:

$$\Box\delta R = 0, \tag{21}$$

where $\Box$ is the d'Alembertian operator.

4.1 Explicit Form of the Wave Operator

The Schwarzschild metric is given by:

$$ds^2 = -f(r)\,dt^2 + \frac{dr^2}{f(r)} + r^2(d\theta^2 + \sin^2\theta\,d\phi^2), \tag{22}$$

with

$$f(r) = 1 - \frac{2M}{r}. \tag{23}$$

The general expression for the d'Alembertian acting on a scalar field ϕ is:

$$\Box\phi = \frac{1}{\sqrt{-g}}\partial_\mu\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\right), \tag{24}$$

where g is the determinant of the metric tensor.

For the Schwarzschild metric, the determinant is:

$$g = \det(g_{\mu\nu}) = -r^4\sin^2\theta, \tag{25}$$

thus

$$\sqrt{-g} = r^2\sin\theta. \tag{26}$$

Substituting into the general formula, the wave operator becomes:

$$\Box\delta R = \frac{1}{r^2\sin\theta}\partial_\mu\left(r^2\sin\theta\,g^{\mu\nu}\partial_\nu\delta R\right). \tag{27}$$

Expanding term-by-term, we find:

$$\Box\delta R = -\frac{1}{f(r)}\partial_t^2\delta R + \frac{1}{r^2}\partial_r\left(r^2 f(r)\partial_r\delta R\right) + \frac{1}{r^2}\Delta_{S^2}\delta R, \tag{28}$$

where Δ_{S^2} is the Laplacian on the unit 2-sphere:

$$\Delta_{S^2} \equiv \frac{1}{\sin\theta}\partial_\theta\left(\sin\theta\,\partial_\theta\right) + \frac{1}{\sin^2\theta}\partial_\phi^2. \tag{29}$$

4.2 Separation of Variables

We propose a separation of variables ansatz for the perturbation:

$$\delta R(t,r,\theta,\phi) = e^{-i\omega t}Y_{\ell m}(\theta,\phi)\frac{\Psi(r)}{r}, \tag{30}$$

where:

- ω is the complex oscillation frequency,
- $Y_{\ell m}(\theta,\phi)$ are the spherical harmonics satisfying

$$\Delta_{S^2}Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}, \tag{31}$$

- $\Psi(r)$ is the radial function to be determined.

Substituting this ansatz into the wave equation will allow us to reduce the problem to an ordinary differential equation (ODE) for $\Psi(r)$.

5 Linearization of the Trace Equation around Schwarzschild Background in the Jordan Frame

In this section, we study the linear perturbations of the trace equation in the Jordan frame, specifically around the Schwarzschild solution. We consider the model $f(R) = R^2$, for which the trace equation reduces to a free massless scalar wave equation:

$$\Box \delta R = 0 \tag{32}$$

where $\Box$ is the D'Alembertian operator constructed using the Schwarzschild background metric. We seek solutions that are separable in time, radial, and angular parts, and we proceed to derive the resulting radial equation governing the perturbations.

5.1 Derivation of the Radial Equation

We begin by assuming a separable ansatz for the perturbation:

$$\delta R(t, r, \theta, \phi) = e^{-i\omega t} Y_{\ell m}(\theta, \phi) \frac{\Psi(r)}{r} \tag{33}$$

where $Y_{\ell m}(\theta, \phi)$ are the standard spherical harmonics.

The Schwarzschild metric is given by:

$$ds^2 = -f(r)\, dt^2 + \frac{dr^2}{f(r)} + r^2 \left(d\theta^2 + \sin^2\theta\, d\phi^2\right) \tag{34}$$

with

$$f(r) = 1 - \frac{2M}{r} \tag{35}$$

Expanding the D'Alembertian operator on a spherically symmetric background, we have:

$$\Box \delta R = -\frac{1}{f(r)} \partial_t^2 \delta R + \frac{1}{r^2} \partial_r \left(r^2 f(r) \partial_r \delta R\right) + \frac{1}{r^2} \Delta_{S^2} \delta R \tag{36}$$

where Δ_{S^2} is the Laplacian on the two-sphere.

Let us compute each term separately:

Time Derivative Term: Using the ansatz,

$$\partial_t^2 \delta R = -\omega^2 e^{-i\omega t} Y_{\ell m} \frac{\Psi(r)}{r} \tag{37}$$

thus,

$$-\frac{1}{f(r)} \partial_t^2 \delta R = \frac{\omega^2}{f(r)} e^{-i\omega t} Y_{\ell m} \frac{\Psi(r)}{r} \tag{38}$$

Angular Laplacian Term: Since spherical harmonics satisfy

$$\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m} \tag{39}$$

we get

$$\frac{1}{r^2} \Delta_{S^2} \delta R = -\frac{\ell(\ell+1)}{r^3} e^{-i\omega t} Y_{\ell m} \Psi(r) \tag{40}$$

Radial Derivative Term: First, compute the radial derivative:

$$\partial_r \delta R = e^{-i\omega t} Y_{\ell m} \left(\frac{\Psi'(r)}{r} - \frac{\Psi(r)}{r^2} \right) \tag{41}$$

thus,

$$r^2 f(r) \partial_r \delta R = e^{-i\omega t} Y_{\ell m} \left(r f(r) \Psi'(r) - f(r)\Psi(r) \right) \tag{42}$$

and further,

$$\partial_r \left(r^2 f(r) \partial_r \delta R \right) = e^{-i\omega t} Y_{\ell m} \left(r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r)\Psi(r) \right) \tag{43}$$

thus,

$$\frac{1}{r^2} \partial_r \left(r^2 f(r) \partial_r \delta R \right) = e^{-i\omega t} Y_{\ell m} \frac{1}{r^2} \left(r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r)\Psi(r) \right) \tag{44}$$

Combining All Terms: Substituting all pieces into the trace equation, we obtain:

$$e^{-i\omega t} Y_{\ell m} \left\{ \frac{\omega^2}{f(r)} \frac{\Psi(r)}{r} + \frac{1}{r^2} \left(r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r)\Psi(r) \right) - \frac{\ell(\ell+1)}{r^3} \Psi(r) \right\} = 0 \tag{45}$$

Since the prefactor $e^{-i\omega t} Y_{\ell m}$ is nonzero, the radial equation simplifies to:

$$\frac{\omega^2}{f(r)} \frac{\Psi(r)}{r} + \frac{1}{r^2} \left[r f'(r) \Psi'(r) + r f(r) \Psi''(r) - f'(r)\Psi(r) \right] - \frac{\ell(\ell+1)}{r^3} \Psi(r) = 0 \tag{46}$$

Summary: In this subsection, we derived the radial perturbation equation for the trace fluctuation δR around a Schwarzschild background in the Jordan frame of $f(R) = R^2$ gravity. The next step will be to recast this radial equation into a Schrödinger-like form to study the quasi-normal modes.

6 Setting up the Modified Regge-Wheeler Equations

Here we introduce the tortoise coordinate and simplify (46), so that it looks like a Schrodinger equation with a potential.

6.1 Tortoise coordinate

The tortoise coordinate r_* is defined by

$$dr_* = \frac{dr}{f(r)} \tag{47}$$

so that

$$\frac{d}{dr} = \frac{1}{f(r)} \frac{d}{dr_*}. \tag{48}$$

Thus,

$$\frac{d^2}{dr^2} = \frac{d}{dr} \left(\frac{d}{dr} \right) = \frac{d}{dr} \left(\frac{1}{f(r)} \frac{d}{dr_*} \right) = -\frac{f'(r)}{f^2(r)} \frac{d}{dr_*} + \frac{1}{f^2(r)} \frac{d^2}{dr_*^2} \tag{49}$$

6.2 Radial Equation in Tortoise Coordinate

Rewrite the radial equation (46), multiplying by r to eliminate denominators as much as possible:

$$\frac{\omega^2}{f(r)}\Psi + \frac{1}{r}\left[rf'(r)\Psi' + rf(r)\Psi'' - f'(r)\Psi\right] - \frac{l(l+1)}{r^2}\Psi = 0$$

Simplify:

$$\omega^2\frac{\Psi}{f(r)} + f'(r)\Psi' + f(r)\Psi'' - \frac{f'(r)}{r}\Psi - \frac{l(l+1)}{r^2}\Psi = 0$$

Now rearranging:

$$f(r)\Psi'' + f'(r)\Psi' + \left(\omega^2\frac{1}{f(r)} - \frac{f'(r)}{r} - \frac{l(l+1)}{r^2}\right)\Psi = 0 \tag{50}$$

Now translate to tortoise coordinate:
Using (48) and (49), (50) can be written as:

$$\frac{d^2\Psi}{dr_*^2} + \left[\omega^2 - V_{\text{eff.}}\right]\Psi = 0, \tag{51}$$

where

$$V_{\text{eff.}} = f(r)\left(\frac{f'(r)}{r} + \frac{l(l+1)}{r^2}\right) \tag{52}$$

is the effective potential.

7 Schrödinger-like Equation and Effective Potential for Scalar Perturbations in $f(R) = R^2$ Gravity

In this section, we proceed with the linearized perturbation theory in $f(R) = R^2$ gravity around the Schwarzschild background. This leads to a form of the radial equation for scalar perturbations, which can be expressed as a Schrödinger-like equation.

7.1 Linearization Around Schwarzschild Background

We begin by considering the Schwarzschild solution for the background spacetime:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\phi^2). \tag{53}$$

For perturbations in the scalar sector, the metric is expressed as:

$$g_{\mu\nu} \to g_{\mu\nu} + h_{\mu\nu}.$$

We then work in the Jordan frame and introduce the tortoise coordinate r_* defined by:

$$dr_* = \frac{dr}{f(r)} \quad \text{with} \quad f(r) = 1 - \frac{2M}{r}.$$

After deriving the equations for the scalar perturbations, we arrive at a Schrödinger-like equation for the radial perturbations of the form:

$$\frac{d^2\Psi}{dr_*^2} + \left[\omega^2 - V_{\text{eff}}(r)\right] \Psi = 0, \tag{54}$$

where the effective potential V_{eff} is given by:

$$V_{\text{eff}} = f(r) \left(\frac{f'(r)}{r} + \frac{l(l+1)}{r^2} \right). \tag{55}$$

7.2 Interpretation of the Effective Potential

The effective potential V_{eff} can be used to analyze the behavior of perturbations to the Ricci scalar in $f(R) = R^2$ gravity. For scalar perturbations in this model, we found that the effective potential takes the form of a **potential well**, as opposed to the potential barrier seen in standard General Relativity for gravitational perturbations.

This may be due to the fact that in $f(R) = R^2$ gravity, we have considered the Schwarzschild metric as the background for the perturbations, which is possible only when we take $R = 0$ and $R_{\mu\nu} = 0$. And just like in GR, there were no propagating scalar modes, here also there are no propagating scalar modes – it is trapped inside the potential well.

7.3 Numerical Solution and Plotting of the Effective Potential

To visualize the effective potential, we used Python to compute and plot V_{eff} as a function of the radial coordinate r for various values of the angular momentum number l. The potential was computed using the formula for V_{eff}, and the maxima of the potential were calculated and marked.

Below is the Python code used to plot the effective potential for various values of l and calculate the positions of the maxima:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import minimize_scalar

# Constants
M = 1.0  # Mass of the black hole in natural units (M = 1)

# Define the effective potential function for scalar perturbations
def V_eff(r, l):
    f_r = 1 - 2 * M / r  # f(r) for Schwarzschild background
    f_prime_r = 2 * M / r**2  # Derivative of f(r)
    return f_r * (f_prime_r / r + l * (l + 1) / r**2)

# Define the range of r values
r_values = np.linspace(2, 20, 1000)  # Radial coordinate range (r > 2M)
l_values = [2, 3, 4, 5]  # Different angular momentum values

# Plot the effective potential for different values of l
plt.figure(figsize=(10, 6))

for l in l_values:
    V_values = [V_eff(r, l) for r in r_values]
    plt.plot(r_values, V_values, label=f'l = {l}')

# Add labels and title
plt.xlabel('r (M)')
plt.ylabel('V_eff(r)')
plt.title('Effective Potential for Scalar Perturbations')
plt.legend()

# Show the plot
plt.grid(True)
plt.show()

# Calculate and print the maxima locations for each l value
for l in l_values:
    def neg_V_eff(r):
        return -V_eff(r, l)

    maxima = minimize_scalar(neg_V_eff, bounds=(2, 20), method='bounded')
    print(f"Maxima for l = {l}: r = {maxima.x:.2f}, V_eff = {V_eff(maxima.x, l):.2f}"
```

7.4 Results and Interpretation

Upon running the Python code, we obtained the following observations:

- The effective potential for scalar perturbations appears as a potential-barrier curve for each value of l.
- The maxima of the potential were located at different radial positions for various l values.

Similar to the Regge-Wheeler potential in General Relativity, which has a potential barrier at a certain value of r, the scalar mode in $f(R)$ gravity also exhibits such behavior.

References

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